Building Fire-Resilience Credit Model

Poisson Initiation with an 8-State Bayesian Containment Chain

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Contents

1 Executive Summary

The Building Fire-Resilience Credit Model (`MKM-FIRE-001`) prices the fire resilience of a commercial asset so that fire loss aggregates into the same resilience-credit currency as the existing 10,000-storm Monte Carlo engine. The credit is consumed by the Commercial PRS Pricer as an independent *full-conflagration* leg, combined with the flood and wind legs through a root-sum-of-squares all-in coupon. The model is built from four governance-separated component models:

- **Model A — Initiation.** A Poisson ignition frequency λ_i per asset, composed from a base annual rate by commercial type and multiplicative modifiers for occupancy, process load, condition, protection and fire history.
- **Model B — Intensity Track.** A 15-minute latent intensity I_t whose growth is damped by passive (structural) defence, scaled by the structural frame's combustibility, cut once active suppression bites, and capped at a combustibility-implied fuel ceiling.
- **Model C — Response Effectiveness.** A deterministic mapping from the asset's CDM resilience levels — and its structural-frame `ConstructionType` — to the dynamic quantities Models B and D consume (detection time, suppression-bite time, growth rate, critical threshold, controllability, the fire-service-reach gate, and the combustibility that drives the growth rate, the fuel ceiling and the passive-containment credit).
- **Model D — Containment, Loss & Resilience Credit.** An 8-state discrete-time Bayesian chain marched in 15-minute steps to an absorbing terminal state, aggregated into the credit currency.

The central concept is the *point of no return* (PNR): the first step at which latent intensity I_t exceeds the critical threshold I_{crit} *before* suppression has bitten — or, equivalently, the first step of any fire in a building where suppression is structurally unreachable *and* the structural frame is combustible. Once PNR latches, the containment chain switches to an absorbing matrix whose Contained column is zero: the fire can no longer be contained and absorbs into {partial loss, total loss}. The PNR frequency, $n_{\text{PNR}}/n_{\text{sim}}$, multiplied by a 100% loss-given-event, is the bps the PRS pricer charges for the fire leg.

Building *material* is a first-class driver: the structural frame's combustibility scales how fast the fire grows (a non-combustible frame buys suppression time), caps the latent intensity below the critical threshold (so a non-combustible frame cannot run to a whole-building conflagration even above the fire-service reach — only a negligible residual remains, effectively zero in practice), and credits the containment chain (a non-combustible structure contains the fire floor-by-floor even when active suppression cannot reach it). This is what stops building height alone from forcing a conflagration in a reinforced-concrete or steel tower.

Status: Development (Tier 2, RAG Amber). All four component models and the end-to-end orchestrator are delivered with unit tests, including the fire-service-reach interaction. Every numeric parameter is a first-pass engineering-judgement seed (`config/fire_matrices.json`, `version "seed-0"`); the results in this document are therefore *directional, not calibrated.*

Key risk: The model uses placeholder calibration throughout. It is not approved for production until historical calibration completes and the Model Risk Committee (MRC) issues sign-off (remediation items FIRE-R1–R8, listed in full in Section 8).

2 Model Purpose and Scope

2.1 Purpose

To assign each commercial asset a fire-resilience credit — expressed as an annual unconditional loss frequency, a conditional containment rate, and a full-conflagration (PNR) frequency — that can be priced pari-passu with the flood and wind hazards already in the portfolio engine. The credit rewards genuine, evidenced fire resilience (detection, suppression, compartmentation, tested procedures) and penalises the configurations that historically drive total losses, most notably tall buildings without adequate internal suppression.

2.2 Scope

The model covers commercial assets described by the v2 commercial-asset and resilience sections of the Common Data Model (CDM). It reads only the derived `AssetFireFeatures` bundle — never the raw CDM record — so that the feature contract is explicit, testable and stable against CDM schema churn. The in-scope drivers are: commercial type, occupancy status, business-rates category, property condition, the nine resilience-level fields (automatic detection, suppression systems, emergency procedures, structural fire resistance, compartments, fire stopping, external materials, access route, business continuity), the structural-frame `ConstructionType`, fire-damage history, and number of storeys. The model produces a per-asset resilience credit and a portfolio roll-up; both are reproducible from a single random seed.

2.3 Out of Scope

- Physical (CFD) fire-spread and thermal-plume modelling.
- Smoke / life-safety egress and casualty modelling.
- Wildland–urban interface and external conflagration spread.
- Business-interruption quantum (only the resilience credit is produced; the loss-given-event is fixed at 100% for the conflagration leg).
- Explicit floor-level geometry — building height enters only through `NumberOfStoreys`.

3 Mathematical Framework

3.1 Initiation (Model A)

The per-asset annual ignition rate is the base rate for the commercial type scaled by five independent multipliers:

$$\lambda_i = \lambda_{0,c}\, m_{\text{occ}}\, m_{\text{proc}}\, m_{\text{cond}}\, m_{\text{protection}}\, m_{\text{history}},$$

where $\lambda_{0,c}$ is the base annual frequency by commercial type c; m_{occ} from occupancy status; m_{proc} from the process load implied by type (raised, never lowered, by a business-rates override such as *Restaurant*); m_{cond} from property condition; $m_{\text{protection}}$ from the mean resilience level across the protection fields; and m_{history} from the severity and recency of any prior fire. Unknown CDM options fall back to a neutral multiplier of 1.0, so a partial record still yields a usable λ_i.

Over a run horizon of h years the effective rate is $\lambda_{\text{eff}} = \lambda_i\, h$. For each of n_{sim} draws a Poisson count $N \sim \text{Poisson}(\lambda_{\text{eff}})$ is sampled, and a fire instantiates for that draw when $N \geq 1$ — the direct analogue of storm-event generation, where only a subset of draws produce an event. Each instantiated fire is tagged with an `InitiationClass` drawn from the per-type categorical prior (Appendix B).

3.2 Response Effectiveness (Model C)

Each resilience level is mapped to an index $k \in \{0, \ldots, 4\}$ in the CDM `RESILIENCE_LEVELS` vocabulary and to a fraction $f = k/4 \in [0, 1]$. A field-group effectiveness is the mean over its fields of the field's configured range linearly interpolated by f:

$$e_{\text{group}} = \frac{1}{|G|} \sum_{j \in G} \left[\, \ell_j + f_j\,(u_j - \ell_j)\,\right],$$

giving the aggregate passive effectiveness e_{passive} (structural resistance, compartments, fire stopping, external materials) and response effectiveness e_{response} (access route, procedures, continuity).

Detection time is $\tau_{\text{det}} = \tau_{\text{det},0}\, \delta_\ell$ where δ_ℓ is the detection multiplier for the asset's detection level. Suppression-bite time applies a gentle per-storey height penalty $p \in [p_{\min}, p_{\max}]$ interpolated over storeys *and* a suppression-level multiplier b_{supp} (a better system engages sooner), but is governed first by the **fire-service-reach gate**: external attack reaches only up to R storeys, and above R effective suppression requires internal sprinklers, i.e. a suppression level index $\geq \theta_{\text{int}}$. Writing s for the storey count and k_{supp} for the suppression-level index,

$$\text{reachable} \;=\; \big(k_{\text{supp}} \geq \theta_{\text{int}}\big) \;\vee\; \big(s \leq R\big),$$

$$\tau_{\text{supp}} = \begin{cases} \tau_{\text{supp},0}\; p\; b_{\text{supp}} & \text{if reachable,} \\ \tau_\infty\; (\gg \text{step budget}) & \text{otherwise.} \end{cases}$$

Compartmentation quality $q = \frac{1}{2}\big(k_{\text{comp}} + k_{\text{stop}}\big)$ selects the well- vs. poorly-compartmented growth rate; the suppression level selects both the post-bite growth multiplier μ_{supp} and the critical intensity I_{crit}, the latter graded across all five levels (not a binary strong/weak step) so every suppression level shifts the controllability race. Finally the controllability score is the weighted blend

$$C = w_p\, e_{\text{passive}} + w_r\, e_{\text{response}} + w_s\, f_{\text{supp}},$$

(renormalised by $w_p + w_r + w_s$), which is remapped to an active matrix-blend weight $w_{\text{active}} = \text{clamp}\big((C - C_{\text{floor}})/(C_{\text{ceil}} - C_{\text{floor}}), 0, 1\big)$. The blend weight that actually mixes the pre-PNR matrices (Model D) takes the larger of the active weight and a *structural-containment* weight $w_{\text{struct}} =$

$\mathrm{clamp}(\kappa\,(1-\gamma), 0, 1)$ driven by the frame combustibility γ (below):

$$w \;=\; \max\bigl(w_{\text{active}},\, w_{\text{struct}}\bigr) \in [0,1].$$

A fire is controllable if it can be *either* actively suppressed *or* passively contained by a non-combustible structure, so a reinforced-concrete frame ($\gamma \to 0$) blends near the controllable matrix even when suppression cannot reach it. A continuous blend, rather than a hard threshold, is what lets the well-graded score C actually grade containment: a hard switch saturates because passive and response defences alone already clear it, leaving suppression unable to move the matrix.

Structural-frame combustibility. The CDM `Construction.ConstructionType` maps to a combustibility fraction $\gamma \in [0,1]$ (0 = non-combustible reinforced concrete; 1 = timber frame; unknown $\to$ a cautious default at the scale midpoint), via `construction.combustibility_by_type`. It drives three quantities the conflagration leg consumes: (i) a *growth scale* interpolated over $[\,g^-, g^+]$ that multiplies the per-step intensity growth, so a non-combustible frame catches and grows slower (the *timing* leg, giving suppression more of the race); (ii) the *intensity ceiling* $I_{\max}$ (Model B); and (iii) the structural-containment weight w_{struct} above. The frame also supplies the structural-fire-resistance fallback level when the granular checklist is silent (a non-combustible frame is the physical evidence for that field), taking priority over the BRI-grade fallback for that one field.

3.3 Intensity Track (Model B)

The latent intensity starts at $I_0 = 0$ and grows each 15-minute step by

$$g_t = g_{\text{base}}\,(1 - w_{\text{damp}}\, e_{\text{passive}}), \qquad g_t \leftarrow g_t \cdot \mu_{\text{supp}} \;\text{ once suppression is active,}$$

$$I_{t+1} = \min\bigl(I_t + g_t,\; I_{\max}\bigr), \qquad g_t \ge 0,$$

where g_{base} is the compartmentation-selected growth rate (already scaled by the combustibility growth factor of Model C) and suppression becomes active at the step $\lceil \tau_{\text{supp}} \rceil$. The *fuel ceiling* $I_{\max}$ is threshold-anchored on combustibility: every non-combustible frame (γ at or below the combustible threshold) shares one low plateau that sits *below* even the worst I_{crit}, so its intensity can never cross the critical threshold via the race; above the threshold the ceiling grades up to a value exceeding any I_{crit} as the frame becomes fully combustible.

The *mean* profile fixes g_{base}, τ_{supp} and $I_{\max}$, but each fire draws its own race realisation: per-fire lognormal jitter is applied once to the growth rate ($g_{\text{base}} \cdot e^{\sigma_g Z}$), the bite time ($\tau_{\text{supp}} \cdot e^{\sigma_\tau Z}$) and the fuel ceiling ($I_{\max} \cdot e^{\sigma_c Z}$) before the march. Without the timing/growth jitter the PNR verdict would be a deterministic property of the asset, forcing the conflagration leg to be all-or-nothing. The ceiling jitter is a structural safeguard against literal immunity: it gives a non-combustible frame with no effective suppression a non-zero chance that an unlucky (high-fuel-load) fire's ceiling exceeds I_{crit} — the "eventually it catches" tail. In practice this residual realises as *effectively zero* (see Table 12), because the chain absorbs before the slowly-creeping capped intensity can cross the threshold (the FIRE-L3 effect); it registers only as a sub-bp tail in very tall non-combustible assets. Setting $\sigma_g = \sigma_\tau = \sigma_c = 0$ recovers the deterministic track and hard cap.

3.4 Point of No Return

PNR latches at the first step t for which suppression is not yet active and

$$I_t > I_{\text{crit}} \quad \vee \quad \big(\neg\,\text{reachable} \,\wedge\, \text{combustible}\big),$$

where *combustible* means the frame combustibility γ clears the combustible threshold; the latch is sticky (once set, it stays set). The second disjunct is the operational heart of the model: a building taller than the fire-service reach with no adequate internal suppression *and a combustible frame* is uncontrollable *from ignition*, so PNR latches at step 1. A *non-combustible* tall, unsprinklered building does *not* auto-latch: its fuel ceiling $I_{\max}$ sits below I_{crit}, so the first disjunct also cannot fire, and the structure contains the fire (only the jittered residual remains). This is what turns building *material* — not height alone — into the live driver of total loss: height forces a conflagration only when the frame is combustible.

3.5 Containment Chain (Model D)

Each fire is a discrete-time chain over the eight states

$$S_0 \text{ Ignition},\; S_1 \text{ Growth},\; S_2 \text{ Detected},\; S_3 \text{ Internal response},\; S_4 \text{ External response},$$

$$S_5 \text{ Contained},\; S_6 \text{ Partial loss},\; S_7 \text{ Total loss},$$

with $\{S_5, S_6, S_7\}$ absorbing. At each step the intensity track advances and the active 8×8 row-stochastic matrix is selected: while PNR is unlatched the pre-PNR matrix is the controllability-weighted blend

$$M \;=\; w\,M_{\text{AA}} \;+\; (1-w)\,M_{\text{weak}},$$

(row-stochastic for any $w \in [0,1]$), interpolating continuously from the weak matrix at $w = 0$ to the controllable matrix at $w = 1$; once PNR latches, every transient state uses the absorbing point-of-no-return matrix, whose S_5 (Contained) column is identically zero. The next state is sampled from the current state's row (with defensive renormalisation) until an absorbing state is reached or the step budget is exhausted. The three base matrices are given in Appendix E.

3.6 Resilience Credit

Aggregating over the n_{sim} draws (of which n_{fires} ignite), with terminal counts $n_{\text{contained}}$, n_{partial}, n_{total} and n_{PNR} fires that reached the point of no return, the credit currency is

$$\text{loss frequency} = \frac{n_{\text{partial}} + n_{\text{total}}}{n_{\text{sim}}}, \qquad \text{containment rate} = \frac{n_{\text{contained}}}{n_{\text{fires}}},$$

$$\text{fire spread (bps)} = \frac{n_{\text{PNR}}}{n_{\text{sim}}} \times 1.0 \times 10{,}000,$$

the last being the full-conflagration leg the Commercial PRS Pricer consumes (frequency $\times$ 100% loss-given-event). It is combined with the flood/wind legs into an all-in coupon by root-sum-of-squares, all-in $= \sqrt{\sum_{\text{perils}} \text{leg}^2}$, treating the perils as independent.

4 Input Parameters and Calibration

All numeric seeds live in `config/fire_matrices.json` (`version "seed-0"`) and are loaded by `config/fire.py`; no number is embedded in the model code. The resilience-level vocabulary is imported from `port.cdm.asset.resil` so the level-to-index-to-fraction mapping stays single-source with the rest of the CDM. The full parameter set — initiation base rates and the five modifier tables, initiation-class priors, the progression dynamics (growth, critical intensity, detection/suppression multipliers, passive/response ranges, height penalty, fire-service reach, timing, controllability weights) and the three transition matrices — is reproduced verbatim in the Appendices.

Calibration status. Every value is a first-pass engineering-judgement seed chosen for plausible *direction* and *ordering*, not magnitude: e.g. hotels and mixed-use ignite more often than offices; worse condition and worse protection raise frequency; better resilience levels shorten detection and suppression times and raise the critical intensity. Calibration against historical commercial fire-loss data (FIRE-R1) and an independent benchmarking exercise (FIRE-R2) are open remediation items; until they close, the outputs must be read as relative resilience signals, not absolute loss probabilities.

5 Implementation Details

5.1 Module Layout

- `config/fire.py` — parameter schema (enums, dataclasses, loader).
- `config/fire_matrices.json` — seed numeric parameters.
- `src/models/fire/data_structures.py` — runtime types (`AssetFireFeatures`, `ResponseProfile`, `IntensityTrack`, `ContainmentOutcome`, `AssetFireResult`).
- `src/models/fire/initiation.py` — Model A (Poisson initiation).
- `src/models/fire/response_effectiveness.py` — Model C.
- `src/models/fire/intensity_track.py` — Model B.
- `src/models/fire/containment.py` — Model D and the end-to-end orchestrator (`simulate_asset_fire`, `simulate_portfolio_fire`).
- `src/routes/commercial/hazard.py` — read-time join that attaches the fire leg (`fire_spread_bps`) to the asset hazard payload.

5.2 Randomness

All randomness flows through a single caller-owned `numpy.random.Generator`, so an entire portfolio run is reproducible from one seed — matching the storm and typhoon model convention. Model C is deterministic, and Model B is deterministic *given a profile*; the per-asset mean profile is drawn into a per-fire realisation inside Model D, which jitters the bite time and growth rate once per fire and then samples the chain march. Every draw from the generator therefore still happens in Model D, and the whole run stays bit-for-bit reproducible.

6 Validation and Backtesting

Historical calibration and backtesting are open remediation items (FIRE-R1, FIRE-R2) — to be completed before production sign-off. Current validation is by directional unit tests, which confirm the model behaves as designed: a strong asset contains far more often and loses far less than a weak one; the point-of-no-return gate is reachable for genuinely weak or tall-and- unsprinklered assets but not for strong ones; the fire-service-reach interaction flips a building from fully containable to a guaranteed conflagration exactly at the reach height when internal suppression is inadequate; and a whole-portfolio run is bit-for-bit reproducible from a single seed. The sensitivity tables in the next section are produced by running the *actual* end-to-end model.

6.1 Automated Test Results

Automated test-results table to be generated. Stage 1 + Stage 2 unit tests live under `tests/models/fire/` (test_initiation.py, test_response_effectiveness.py, test_intensity_track.py, test_containment.py).

7 Sensitivity Analysis

The tables below are generated by `docs/models/sensitivities/fire_resilience/` running the real `simulate_asset_fire` against a fixed baseline (a fully-occupied 3-storey *Office* in *Fair* condition with every resilience field at *Meets minimum*), varying one input at a time over 20,000 Monte Carlo draws per cell.

The results show a clear **division of labour** across the four levers, which is how the credit is designed to respond.

Suppression drives containment and loss frequency. On the suppression sweep the containment rate climbs monotonically from ≈ 47% at *Not assessed* to ≈ 87% at *Verified*, and the annual loss frequency falls roughly four-fold (1.15% → 0.26%). This is the continuous matrix blend at work: a better-suppressed asset has a higher controllability score C, a blend weight w nearer the controllable matrix, and so a higher chance of reaching S_5 (Contained). This is the economically meaningful credit the suppression system buys.

Compartmentation drives conflagration. Passive compartmentation selects the growth rate, so a poorly-compartmented asset grows fast enough to cross I_{crit} before the chain absorbs and shows a small but non-zero fire-spread (≈ 5 bps), whereas a well-compartmented one stays at zero. Commercial type and condition move loss frequency in the expected order (hotels, mixed-use and poor condition worst).

The fire-service reach drives the catastrophic tail. The height sweep (held at sub-threshold *Partial* internal suppression *and* the default combustibility) is the headline non-linearity: at or below the 4-storey reach the credit is ordinary (≈ 63% containment, zero spread), but at *five storeys and above* suppression can no longer engage and a combustible-frame asset collapses to 0% containment and ≈ 203 bps of fire spread — a **cliff, not a gradient**. The height × suppression grid confirms the cliff appears precisely when internal suppression is below the sprinkler threshold; *Meets minimum* or better keeps the asset reachable at any height.

Building material gates the high-rise tail. The construction sweep (a 12-storey, sub-threshold-suppressed asset — above the reach, so material is the only thing left to govern the outcome) is the new lever. A non-combustible frame is contained by its own structure: *reinforced concrete* ≈ 88% containment with near-zero spread, *steel / brick* ≈ 79%, *modern methods* ≈ 64% (seeded less combustible than mixed), while *mixed construction* and *timber* stay combustible and suffer the full 0%/≈ 203 bps conflagration. The reach cliff above therefore fires only for combustible frames; material, not height alone, decides the high-rise tail.

Supporting tables. Six further tables isolate the new mechanisms and the config. The **construction × height** grid (Table 10) confirms the reach cliff fires *only* for combustible frames: a reinforced-concrete or steel frame holds zero spread at 3, 5, 8 and 20 storeys, while mixed and timber jump to ≈ 203 bps above the reach. The **construction × suppression** grid (Table 11) shows the passive and active credits are complementary: a non-combustible frame holds ≈ 88% containment at *every* suppression level, whereas a combustible frame climbs from 0% (sub-threshold, unreachable) to ≈ 87% once internal suppression clears the sprinkler threshold. The **detection** sweep (Table 8) is deliberately near-flat — detection only shifts the bite timing, and the chain absorbs before it matters (the same FIRE-L3 effect that mutes the conflagration leg); the **occupancy** sweep (Table 9) moves loss frequency the expected small amount (a vacant building ignites less). The **no-suppression robustness check** (Table 12) confirms that even with suppression absent a non-combustible frame realises effectively zero conflagration — the fuel cap is robust. Finally the deterministic **config explainer** (Table 13) maps each `ConstructionType` to the five model levers it sets (γ, growth scale, $I_{\max}$, w_{struct}, and the structural-fire-resistance fallback), computed directly from `config/fire_matrices.json` so the table cannot drift from the running model.

Table 1: Fire-resilience credit by suppression-systems level (3-storey office baseline)

	Containment (%)	Loss freq (%)	Fire spread (bps)
Suppression: Not assessed	47.0	1.15	8.50
Suppression: Partial	63.1	0.7500	0.0000
Suppression: Meets minimum	66.7	0.6750	0.0000
Suppression: Enhanced	78.1	0.4450	0.0000
Suppression: Verified	86.8	0.2600	0.0000

Table 2: Fire-resilience credit by building height — the fire-service reach cliff (reach = 4 storeys; sub-threshold 'Partial' internal suppression)

	Containment (%)	Loss freq (%)	Fire spread (bps)
1 storeys	63.1	0.7500	0.0000
3 storeys	63.1	0.7500	0.0000
4 storeys	63.1	0.7500	0.0000
5 storeys	0.0000	2.03	203.0
8 storeys	0.0000	2.03	203.0
12 storeys	0.0000	2.03	203.0
20 storeys	0.0000	2.03	203.0

Table 3: Fire-resilience credit by compartmentation level

	Containment (%)	Loss freq (%)	Fire spread (bps)
Compartments: Not assessed	63.5	0.7400	4.50
Compartments: Partial	63.5	0.7400	5.00
Compartments: Meets minimum	66.7	0.6750	0.0000
Compartments: Enhanced	66.0	0.6900	0.0000
Compartments: Verified	66.3	0.6850	0.0000

Table 4: Fire-resilience credit by property condition

	Containment (%)	Loss freq (%)	Fire spread (bps)
Excellent	68.3	0.5250	0.0000
Good	67.0	0.6250	0.0000
Fair	66.7	0.6750	0.0000
Poor	66.5	0.7900	0.0000
Very poor	64.6	0.9700	0.0000

Table 5: Fire-resilience credit by commercial type

	Containment (%)	Loss freq (%)	Fire spread (bps)
Office	66.7	0.6750	0.0000
Retail	64.6	1.19	0.0000
Hotel	66.8	1.43	0.0000
Leisure	66.4	1.25	0.0000
Healthcare	65.4	0.9750	0.0000
MultiFamily	64.0	1.16	0.0000
MixedUse	66.4	1.52	0.0000
Other	66.4	1.52	0.0000

Table 6: Fire spread (bps) by height × suppression level — the fire-service-reach interaction

	Partial	Meets minimum	Enhanced	Verified
3 storeys	0.0000	0.0000	0.0000	0.0000
5 storeys	203.0	0.0000	0.0000	0.0000
8 storeys	203.0	0.0000	0.0000	0.0000
20 storeys	203.0	0.0000	0.0000	0.0000

Table 7: Fire-resilience credit by construction type / structural-frame combustibility (12-storey, sub-threshold 'Partial' suppression — above the fire-service reach, so material governs)

	Containment (%)	Loss freq (%)	Fire spread (bps)
Reinforced concrete	87.7	0.2500	0.0000
Concrete frame	83.5	0.3350	0.0000
Steel frame	78.8	0.4300	0.0000
Brick and block	78.8	0.4300	0.0000
Modern methods	63.8	0.7350	0.0000
Mixed construction	0.0000	2.03	203.0
Timber frame	0.0000	2.03	203.0

Table 8: Fire-resilience credit by automatic-detection level

	Containment (%)	Loss freq (%)	Fire spread (bps)
Detection: Not assessed	66.9	0.7150	0.0000
Detection: Partial	66.7	0.6750	0.0000
Detection: Meets minimum	66.7	0.6750	0.0000
Detection: Enhanced	66.7	0.6750	0.0000
Detection: Verified	67.3	0.6450	0.0000

Table 9: Fire-resilience credit by occupancy status (a Model A initiation-frequency driver)

	Containment (%)	Loss freq (%)	Fire spread (bps)
Fully occupied	66.7	0.6750	0.0000
Partially vacant	67.3	0.6150	0.0000
Vacant	67.4	0.5750	0.0000

Table 10: Fire spread (bps) by construction × height (sub-threshold 'Partial' suppression) — material, not height alone, gates the conflagration: non-combustible frames stay near zero at every height; combustible frames hit the reach cliff

	3 st	5 st	8 st	20 st
Reinforced concrete	0.0000	0.0000	0.0000	0.0000
Steel frame	0.0000	0.0000	0.0000	0.0000
Modern methods	0.0000	0.0000	0.0000	0.0000
Mixed construction	0.0000	203.0	203.0	203.0
Timber frame	3.50	203.0	203.0	203.0

Table 11: Containment (%) by construction × suppression level (12-storey, above the reach) — a non-combustible frame is contained by its structure even when suppression is weak; the two credits are complementary

	Partial	Meets minimum	Enhanced	Verified
Reinforced concrete	87.7	87.7	87.7	87.1
Steel frame	78.8	78.8	78.8	86.8
Modern methods	63.8	66.7	78.1	86.8
Mixed construction	0.0000	66.7	78.1	86.8
Timber frame	0.0000	66.5	78.1	86.8

Table 12: Construction sweep at *no* effective suppression (12-storey, 'Not assessed') — the robustness check. Even with suppression absent, a non-combustible frame realises effectively zero conflagration: the fuel cap holds because the chain absorbs before the capped intensity can creep across I_{crit} (FIRE-L3). The ceiling jitter leaves only a negligible tail (sub-bp at this baseline; single-digit bps in very tall non-combustible assets). Combustible frames suffer the full cliff.

	Containment (%)	Loss freq (%)	Fire spread (bps)
Reinforced concrete	87.5	0.2700	0.0000
Concrete frame	83.3	0.3600	0.0000
Steel frame	78.7	0.4600	0.0000
Brick and block	78.7	0.4600	0.0000
Modern methods	63.9	0.7800	0.0000
Mixed construction	0.0000	2.16	216.0
Timber frame	0.0000	2.16	216.0

Table 13: Config explainer: each `ConstructionType` maps to the model levers it sets — frame combustibility γ, the growth-rate scale, the fuel ceiling $I_{\max}$, the structural-containment blend floor w_{struct}, and the structural-fire-resistance fallback level. Deterministic (computed from `config/fire_matrices.json`).

Construction	γ	Growth scale	$I_{\max}$	w_{struct}	Struct. FR fallback
Reinforced concrete	0.0000	0.5000	5.00	1.00	Verified
Concrete frame	0.0500	0.5500	5.00	0.9500	Verified
Steel frame	0.1000	0.6000	5.00	0.9000	Enhanced
Brick and block	0.1000	0.6000	5.00	0.9000	Enhanced
Modern methods	0.2500	0.7500	5.00	0.7500	Meets minimum
Mixed construction	0.5000	1.00	37.9	0.5000	Partial
Timber frame	1.00	1.50	120.0	0.0000	Partial

7.1 Interpretation and known model weaknesses

The sensitivity surface above is directionally correct, but two structural features of the seed model deserve to be read alongside it:

- **The conflagration leg is muted for reachable buildings (by construction).** The chain march absorbs in ≈ 3 steps on average, while latent intensity climbs only ≈ 0.8 per step, so a *reachable* fire is contained or lost long before the slow point-of-no-return race can cross I_{crit}. The PNR gate therefore fires almost only for the structural reach cliff (tall, under-sprinklered) plus a sliver of no-suppression cases. This is *semantically* correct — a fire contained in three steps did not reach a point of no return — but it means fire-spread (bps) is driven chiefly by compartmentation and the reach cliff rather than by the active suppression level. Making spread respond strongly to suppression as well would require retuning the intensity-race timescale (a lower I_{crit}, of order 2–4) so the race competes with the three-step march; that is deferred as an enhancement because it is arguably less physical than letting compartmentation own conflagration.

- **The height sweep only shows the cliff at sub-threshold internal suppression.** A building with *Meets minimum* or better internal sprinklers is reachable at any height, so its height sweep is flat by design; the cliff table here deliberately holds suppression at *Partial* to expose the reach behaviour. A future enhancement would model fire-service reach as a brigade-specific, possibly height-graded capability rather than a single global 4-storey threshold, and would let internal-suppression adequacy degrade with height rather than flip at a hard index.

The exposed calibration knobs make the surface tunable without code changes: $C_{\text{floor}}/C_{\text{ceil}}$ position and stretch the containment band; `suppression_bite_multiplier_by_level` and `i_crit_by_level` set the gradient steepness; the `race_jitter` sigmas ($\sigma_\tau, \sigma_g, \sigma_c$) control how smoothly the conflagration probability grades and how large the non-combustible residual tail is; and the `construction` block (`combustibility_by_type`, `combustible_threshold`, `intensity_ceiling`, `growth_combustibility_scale`, `structural_containment_weight`) sets where each frame material sits on the combustibility axis and how strongly it gates growth, the fuel cap and passive containment.

8 Model Limitations and Known Weaknesses

- **No historical calibration (FIRE-L1).** Every parameter is a first-pass placeholder; outputs are relative signals, not absolute loss probabilities.
- **Height proxy (FIRE-L2).** Building height is proxied by `NumberOfStoreys`; the commercial schema carries no explicit floor-level or floor-area metric, and the fire-service reach is a single global threshold rather than a brigade-specific capability.
- **Fast chain absorption mutes the time-dependent dynamics (FIRE-L3).** The chain march absorbs in ≈ 3 steps — faster than the latent intensity race reaches I_{crit} — and this single structural feature mutes three sensitivities that one would expect to be live, each visible in the Section 7 tables and stated here without flattery:
  - *The conflagration leg is insensitive to active suppression.* Fire-spread (bps) responds chiefly to compartmentation and the reach cliff, not the suppression level; suppression instead expresses itself through containment rate and loss frequency.
  - *Automatic detection is near-flat* (Table 8): containment moves < 1 percentage point across all five detection levels, because detection only shifts the bite timing and the chain has usually absorbed before that matters. The detection credit is therefore currently immaterial.

  - *The non-combustible residual realises as effectively zero* (Table 12): even with the ceiling jitter and no suppression, the capped intensity cannot creep across I_{crit} before the chain absorbs, so the "eventually it catches" tail is sub-bp rather than a graded residual.

  Retuning the intensity-race timescale (e.g. a lower I_{crit} of order 2–4 so the race competes with the three-step march) so that suppression, detection and the residual all become live is an open remediation item (FIRE-R5).

- **Single global reach threshold (FIRE-L4).** The fire-service reach is one global 4-storey cut-off and internal-suppression adequacy flips at a hard index, so the height cliff only appears at sub-threshold internal suppression. A brigade-specific, height-graded reach and a graded internal-suppression adequacy are an open enhancement (FIRE-R6).

- **Initiation class is cosmetic.** The initiation-class draw seeds an entry-point label but does not yet change the chain start state or dynamics.

- **Fixed loss-given-event; no partial-loss quantum (FIRE-L6).** The conflagration leg prices *every* point-of-no-return event at a 100% loss-given-event — including the fires the absorbing matrix sends to *partial* loss (S_6), not total loss (S_7). Because n_{PNR} counts partial- and total-loss fires alike but is priced at full total-loss severity, the fire-spread bps *overstates* a true severity-weighted leg. Replacing the fixed LGE with a modelled partial/total severity split is an open remediation item (FIRE-R8); it is the most material conservative bias in the leg after calibration.

- **Construction combustibility is a single seeded fraction (FIRE-L5).** Frame combustibility is one number per `ConstructionType`; the threshold separating non-combustible from combustible frames, the placement of the ambiguous categories (*Modern methods* seeded below the threshold and less combustible than *Mixed construction*), the growth-scale band and the ceiling-jitter sigma are engineering-judgement seeds, not calibrated. Combustibility is a property of the structural frame only — combustible cladding over a non-combustible frame (the Grenfell case) is not yet modelled. Recalibration against loss experience is an open item (FIRE-R7).

- **Independence assumption.** The root-sum-of-squares all-in coupon treats fire, flood and wind as independent perils.

8.1 Outstanding remediation

The model is gated to Development status; MRC sign-off is contingent on closing the items below. They are listed in full — including the ones that expose the model's current weaknesses — so the outstanding work is transparent. Items FIRE-R5–R8 each map to a documented limitation above.

ID	Priority	Item
FIRE-R1	High	Replace the placeholder seed calibration in `fire_matrices.json` with values derived from commercial fire-incident / loss data (closes FIRE-L1).
FIRE-R2	High	Independent benchmarking of the credit against an external fire-loss reference.
FIRE-R3	Medium	Define the production monitoring plan (drift in input distributions; backtest of realised vs. predicted fire frequency).
FIRE-R4	High	Independent model validation and MRC review / sign-off.
FIRE-R5	Medium	Retune the intensity-race timescale (lower I_{crit} so the race competes with the $\approx$ 3-step chain absorption) so that the conflagration leg, the automatic-detection sensitivity and the non-combustible residual all become live rather than muted (closes FIRE-L3).
FIRE-R6	Low	Replace the single global 4-storey fire-service reach with a brigade-specific, height-graded capability and a graded internal-suppression adequacy (closes FIRE-L4).
FIRE-R7	Medium	Recalibrate the construction combustibility seeds (`combustibility_by_type`, `combustible_threshold`, `growth_combustibility_scale`, σ_c) against loss experience and add a combustible-cladding-over-frame input (closes FIRE-L5).
FIRE-R8	Medium	Replace the fixed 100% loss-given-event with a modelled partial/total severity split, so the fire-spread leg is severity-weighted rather than counting every point-of-no-return fire as a total loss (closes FIRE-L6).

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Model purpose and use

"What is the model used for, and is that use consistent with its design?"

The model produces a fire-resilience credit (loss frequency, containment rate and a full-conflagration bps leg) consumed by the Commercial PRS Pricer as an independent peril. Use is consistent with design; it is *not* a life-safety or absolute-loss model and must not be used as one.

9.2 Q2 — Data quality and lineage

"Are the model inputs of sufficient quality, and is their lineage documented?"

Inputs are the CDM-derived `AssetFireFeatures`; lineage runs from the v2 CDM resilience fields through Model C. Missing options degrade gracefully to neutral multipliers. Data *quality* for calibration (historical fire-loss records) is not yet established — open under FIRE-R1.

9.3 Q3 — Conceptual soundness

"Is the methodology theoretically sound and appropriate?"

The four-model decomposition (Poisson initiation, per-fire-stochastic intensity track, resilience→dynamics mapping, controllability-blended absorbing Bayesian chain) is a standard and defensible structure. The point-of-no-return gate is the one bespoke mechanism and is documented in Section 3.4. Its known weakness — that the gate is muted for reachable buildings because the chain absorbs faster than the intensity race — is documented in Sections 7 and 8 (FIRE-L3).

9.4 Q4 — Calibration and parameters

"How were parameters estimated, and are they appropriate?"

All parameters are engineering-judgement seeds (`version "seed-0"`) chosen for direction and ordering, not magnitude. Statistical calibration is an open remediation item (FIRE-R1, FIRE-R2). This is the dominant model limitation.

9.5 Q5 — Implementation verification

"Has the implementation been verified against the specification?"

Yes, by unit tests covering each component model, the orchestrator and the fire-service-reach interaction, plus a reproducibility test. The sensitivity harness exercises the full end-to-end path.

9.6 Q6 — Sensitivity and stability

"How sensitive are outputs to inputs and assumptions?"

See Section 7. Outputs respond in the expected order to type, condition and resilience levels, and exhibit the designed height×suppression cliff. Stability under draw count is adequate at $n_{\mathrm{sim}} = 20{,}000$; formal stability bands are an open item.

9.7 Q7 — Limitations and weaknesses

"Are the model's limitations understood and documented?"

Yes — see Section 8 (FIRE-L1–L6 plus two further documented assumptions), the consolidated remediation table (FIRE-R1–R8), and the "Interpretation and known model weaknesses" discussion in Section 7. The findings that do *not* flatter the model are stated explicitly and tied to remediation: the fast chain absorption that mutes the conflagration leg, the automatic-detection sensitivity and the non-combustible residual (FIRE-L3 → FIRE-R5), and the fixed 100% loss-given-event that overstates the leg by counting partial losses as total (FIRE-L6 → FIRE-R8). The placeholder calibration (FIRE-L1) and the height proxy (FIRE-L2) remain the most material.

9.8 Q8 — Ongoing monitoring

"How will the model be monitored in production?"

Not yet in production. A monitoring plan (drift in input distributions, backtest of realised vs. predicted fire frequency) will be defined as part of FIRE-R3 before sign-off.

9.9 Q9 — Governance and approval

"What is the approval status and who owns the model?"

Owner: David K Kelly. Status: Development, Tier 2, RAG Amber. Not approved for production; MRC sign-off is contingent on closing FIRE-R1–R8 (Section 8).

10 Change History

Version	Date	Description
0.01	05-June-2026	Skeleton document. Stage 1 (Model A) and Stage 2 (Models B/C/D + orchestrator) delivered with unit tests; placeholder calibration.
0.20	06-June-2026	Completed all narrative sections; added the full mathematical framework (Models A–D, PNR, credit currency), governance validation questions, the matrices appendix, and the end-to-end sensitivity analysis.
0.30	08-June-2026	Recalibrated the resilience response: replaced the saturated hard controllability switch with a continuous matrix blend ($M = wM_{\mathrm{AA}} + (1-w)M_{\mathrm{weak}}$); graded I_{crit} across all five suppression levels; added a per-level suppression-bite multiplier and per-fire race jitter (σ_τ, σ_g) so the conflagration probability is no longer all-or-nothing. Containment now grades 47%→88% across suppression levels (previously flat ≈ 88.7%). Reworked the height sweep to expose the 4-storey reach cliff and documented the muted-conflagration and single-reach-threshold weaknesses (FIRE-L3/L4, FIRE-R5/R6).
0.40	16-June-2026	Wired the structural-frame `ConstructionType` into the model as building-material sensitivity: a combustibility fraction γ now (i) scales the per-step growth (timing leg), (ii) caps the latent intensity at a threshold-anchored fuel ceiling $I_{\max}$ with a per-fire ceiling jitter σ_c as a structural residual safeguard (realising ≈ 0 in practice), and (iii) floors the pre-PNR blend weight at $\kappa(1-\gamma)$ (passive structural containment). The unreachable-suppression auto-PNR latch is now gated by combustibility, and `ConstructionType` supplies the structural-fire-resistance fallback. Fixes commercial conflagration being ∼10× too high and material-insensitive (a non-combustible high-rise no longer runs to total loss on every fire); added the construction sensitivity table and limitation FIRE-L5 / remediation FIRE-R7. Expanded the sensitivity analysis to thirteen tables (construction×height and construction×suppression grids, detection, occupancy, a no-suppression robustness check and a deterministic config explainer). Made the unflattering findings explicit and tied them to remediation: fast chain absorption mutes the conflagration leg, the detection sensitivity and the non-combustible residual (FIRE-L3→R5), and the fixed 100% loss-given-event overstates the leg by pricing partial losses as total (FIRE-L6→R8); added a consolidated FIRE-R1–R8 remediation table. Also unified the port Monte Carlo draw count into a single `--num-sims` flag (default 10,000) across storm/fire/seismic.

A Initiation Parameters (Model A)

Table 14: Base annual ignition frequency $\lambda_{0,c}$ by commercial type.

Commercial type	$\lambda_{0,c}$
Office	0.020
Retail	0.030
Hotel	0.035
Leisure	0.030
Healthcare	0.025
MultiFamily	0.030
MixedUse	0.040
Other	0.040

Table 15: Frequency modifiers. Process load is by type with a business-rates override (*Restaurant* = 1.25) that only ever raises the value.

Modifier	Option	Value
m_{occ} (occupancy)	Fully occupied	1.00
	Partially vacant	0.90
	Vacant	0.85
m_{proc} (process)	Office	1.00
	Retail	1.10
	Hotel	1.25
	Leisure	1.25
	Healthcare	1.10
	MultiFamily	1.05
	MixedUse	1.15
	Other	1.15
m_{cond} (condition)	Excellent	0.85
	Good	0.95
	Fair	1.05
	Poor	1.20
	Very poor	1.40
$m_{\mathrm{protection}}$ (level)	Not assessed	1.10
	Partial	1.00
	Meets minimum	0.95
	Enhanced	0.90
	Verified	0.85
m_{history}	No prior	1.00
	Old minor	1.05
	Recent minor	1.10
	Recent moderate/severe	1.40

A fire counts as *recent* if it occurred within `recent_years` = 5 years.

B Initiation-Class Priors

Table 16: Categorical priors over initiation class by commercial type (IE = InternalElectrical, KO = KitchenOccupant, PT = PlantTechnical, PR = ProcessRelated, BS = BasementService, EE = ExternalExposure, Oth = Other). Rows sum to 1.

Type	IE	KO	PT	PR	BS	EE	Oth
Office	0.35	0.10	0.15	0.05	0.15	0.10	0.10
Retail	0.30	0.30	0.10	0.05	0.10	0.10	0.05
Hotel	0.28	0.25	0.15	0.05	0.12	0.10	0.05
Leisure	0.30	0.22	0.13	0.05	0.10	0.10	0.10
Healthcare	0.32	0.12	0.20	0.06	0.12	0.08	0.10
MultiFamily	0.35	0.30	0.08	0.02	0.10	0.08	0.07
MixedUse	0.28	0.12	0.15	0.12	0.10	0.08	0.15
Other	0.28	0.12	0.15	0.12	0.10	0.08	0.15

C Progression Dynamics (Models B & C)

Table 17: Growth, critical intensity, and per-level detection / suppression multipliers.

Block	Key	Value
Growth per step	well_compartmented	1.5
	poorly_compartmented	4.0
I_{crit} by suppression level	Not assessed	6
	Partial	10
	Meets minimum	16
	Enhanced	30
	Verified	55
Detection mult. δ_ℓ	Not assessed	1.00
	Partial	0.90
	Meets minimum	0.80
	Enhanced	0.72
	Verified	0.65
Suppression growth mult. μ_{supp}	Not assessed	1.00
	Partial	0.85
	Meets minimum	0.75
	Enhanced	0.62
	Verified	0.50
Suppression bite mult. b_{supp} (lower = sooner)	Not assessed	3.00
	Partial	2.00
	Meets minimum	1.30
	Enhanced	0.85
	Verified	0.55

Table 18: Passive and response effectiveness ranges $[\ell, u]$, interpolated by the field's resilience-level fraction.

Group	Field	ℓ	u
Passive	StructuralFireResistanceAdequate	0.60	0.95
	CompartmentsProvided	0.55	0.90
	FireStoppingAtPenetrations	0.70	0.95
	ExternalMaterialsFireResistant	0.65	0.95
Response	AccessRouteResilient	0.70	0.95
	EmergencyProceduresTested	0.80	0.97
	BusinessContinuityPlanInPlace	0.85	0.98

Table 19: Height penalty, fire-service reach, timing, intensity dynamics and controllability.

Block	Key	Value
Height penalty	range	[1.05, 1.60]
	max_storeys	20
Fire-service reach	reach_storeys R	4
	internal_suppression_threshold θ_{int}	2
	no_reach_bite_steps τ_∞	999.0
Timing	detection_base_steps $\tau_{\text{det},0}$	2.0
	suppression_bite_base_steps $\tau_{\text{supp},0}$	3.0
Intensity dynamics	passive_damp_weight w_{damp}	0.60
	compartmentation_well_threshold	2.0
Race jitter	timing_sigma σ_τ	0.50
	growth_sigma σ_g	0.40
	ceiling_sigma σ_c	0.10
Controllability	weight (passive) w_p	0.35
	weight (response) w_r	0.35
	weight (suppression) w_s	0.30
	blend_floor C_{floor}	0.18
	blend_ceiling C_{ceil}	0.88

D Structural-Frame Combustibility (Construction)

Table 20: Combustibility fraction γ by `ConstructionType` (0 = non-combustible, 1 = fully combustible). Frames at or below the combustible threshold ($\gamma \leq 0.30$) cannot conflagrate via the intensity race (their fuel ceiling sits below the worst I_{crit}); only the jittered residual remains. An unknown type uses the default 0.50.

ConstructionType	γ
Reinforced concrete	0.00
Concrete frame	0.05
Steel frame	0.10
Brick and block	0.10
Masonry / Stone	0.12
Modern methods	0.25
Mixed construction	0.50
Timber frame	1.00
default (unknown)	0.50

Table 21: Combustibility-driven dynamics in the `construction` block.

Block	Key	Value
Threshold	combustible_threshold	0.30
Fuel ceiling	intensity_ceiling [plateau, max]	[5.0, 120.0]
Growth scale	growth_combustibility_scale $[g^-, g^+]$	[0.5, 1.5]
Containment	structural_containment_weight κ	1.0

The fuel ceiling is threshold-anchored: every frame with $\gamma \leq 0.30$ shares the plateau 5.0 (below the worst $I_{\text{crit}} = 6$), and above the threshold the ceiling grades linearly to 120 at $\gamma = 1$. The growth scale is interpolated by γ and is neutral (1.0) at the midpoint, so a record with no construction type is unchanged. The structural-containment weight floors the pre-PNR blend weight at $\kappa(1 - \gamma)$.

E Containment Transition Matrices (Model D)

Each is an 8×8 row-stochastic matrix over $\{S_0, \ldots, S_7\}$; the step length is 15 minutes. The point-of-no-return matrix has a zero S_5 (Contained) column — once latched, containment is unreachable.

Table 22: `AA_controllable` — pre-PNR controllable endpoint; the active matrix at blend weight $w = 1$ (controllability $C \geq C_{\text{ceil}}$).

From\To	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
S_0	0.05	0.25	0.70	0	0	0	0	0
S_1	0	0.10	0.85	0	0	0	0.05	0
S_2	0	0	0	0.90	0.08	0	0.02	0
S_3	0	0	0	0	0.10	0.85	0.05	0
S_4	0	0	0	0	0	0.75	0.20	0.05
S_5	0	0	0	0	0	1.00	0	0
S_6	0	0	0	0	0	0	0.85	0.15
S_7	0	0	0	0	0	0	0	1.00

Table 23: `weak_controllable` — pre-PNR weak endpoint; the active matrix at blend weight $w = 0$ (controllability $C \leq C_{\text{floor}}$). For intermediate C the active matrix is $w\, M_{\text{AA}} + (1 - w)\, M_{\text{weak}}$.

From\To	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
S_0	0.05	0.70	0.25	0	0	0	0	0
S_1	0	0.10	0.45	0	0	0	0.45	0
S_2	0	0	0	0.40	0.40	0	0.20	0
S_3	0	0	0	0	0.15	0.15	0.70	0
S_4	0	0	0	0	0	0.25	0.50	0.25
S_5	0	0	0	0	0	1.00	0	0
S_6	0	0	0	0	0	0	0.50	0.50
S_7	0	0	0	0	0	0	0	1.00

Table 24: `point_of_no_return` — absorbing, post-PNR. The S_5 column is identically zero.

From\To	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
S_0	0	1.00	0	0	0	0	0	0
S_1	0	0	0	0	0	0	0.40	0.60
S_2	0	0	0	0	0	0	0.40	0.60
S_3	0	0	0	0	0	0	0.40	0.60
S_4	0	0	0	0	0	0	0.40	0.60
S_5	0	0	0	0	0	1.00	0	0
S_6	0	0	0	0	0	0	0.35	0.65
S_7	0	0	0	0	0	0	0	1.00